

Antarctic Science Platform: Community Data Principles (draft)

Definitions:

Data: A set of values, symbols, or signs (recorded on any type of medium) that describe one or more properties of an entity. For example, the numbers generated by a sensor, values derived from a model or analysis, text entered into a survey, audio or video files, or the raw text of a document. Generally speaking, data are used to quantitatively or qualitatively describe one or more persons, objects and systems. Research data provide the evidence base for supporting or refuting ideas in a scientific manner¹.

Metadata: Metadata provide information that describe the data source, and the time, place, and conditions under which the data were created. Metadata inform the user of who, when, what, where, why, and how data were generated. Metadata allow the data to be traced to a known origin and known quality. Metadata can be used for discovery and identification of data collections, to provide information on structural aspects of the data, and to provide administrative information on aspects such as ownership and licensing¹.

Background

New Zealand's Antarctic and Southern Ocean data collection is widely fragmented across numerous repositories and institutions. To address this issue the Antarctic Science Platform in collaboration with Antarctica New Zealand, has established a metadata catalogue to provide a centralised portal for discovery and access to Antarctic and Southern Ocean data collected by New Zealand based researchers (<https://antcat.antarcticanz.govt.nz>). The catalogue is a distributed metadata management system which creates, harvests and replicates metadata from other sources – the catalogue does not currently host data but provides direct access to the data when a data link is available. This creates a virtual data centre providing seamless data discovery, access, and usage across the domain specific and physically distributed data repositories.

Purpose

This document provides seven guiding principles which are intended to promote best practice and support the underlying design and functionality of the metadata catalogue. The principles cover what data providers should consider when evaluating what attributes of their data need to be formatted and described now, and how this might influence the utility of their data in future.

International Context

The catalogue is the most comprehensive database of New Zealand based researchers' Antarctic and Southern Ocean data. The system is designed to disseminate its data to NASA's Antarctic Metadata Directory ([AMD](#)) and other domain specific catalogues/tools/products (e.g. [EMODnet/SOOSmap](#)). This approach is consistent with data centers at other polar programmes (e.g. USAP, BAS, AADC), contributing to the free access and exchange of observations and results from Antarctica^{2,3}.

Implementation

The principles are not intended to be prescriptive, and acknowledge that some Antarctic dataset have limited reuse characteristics which could make implementation of these principles impractical. (e.g. standardised formats have not been developed for certain data; see principle: '*Data are Standardised*', or data are analytical outputs, the data are too large to store uncompressed (see principle: '*Data are Accessible*'), or data are sensitive (see principle: '*CARE Principles for Indigenous Data Governance*'). Data providers who incorporate these principles into their data archival processes will ensure their data are fully incorporated into the Antarctic Metadata Catalogue System, and data described with metadata will be incorporated into the Antarctic Metadata Catalogue (AntCat). At a minimum, data are best described with an internationally recognised metadata standard which will enable machine

interoperability and the sharing of data across other polar data systems (e.g. [Polar Data Search, AMD](#)). Data described with a controlled keyword vocabulary (*see principles: 'Metadata' and 'Data are Defined with Controlled Vocabularies'*) improve findability and interoperability by enabling related metadata and data to be linked by their data's characteristics (e.g. parameters, instruments, projects). Datasets need to be URL accessible in that a single dataset can be downloaded with a distinct URL (see principle: 'Data are Accessible').

Principles

Name	Statement	Context
Metadata ⁴	Data are best represented by metadata described with an internationally recognised standard.	• Carry attributes/fields that enable discovery, access, context, and reuse. • Easily exchanged across data repositories and metadata catalogues and indexed with federated search tools.
Data Granularity	Data are best described and archived in their most granular, usable form most appropriate to the discipline and data type.	• Certain disciplines and data types are best represented by metadata at the project level, while the utility of other data is improved when described and archived at more granular levels.
Data are Standardised	Standardised data formats in widespread use worldwide increase data utility.	• Promote data system interoperability. • Simplify aggregation across multiple data sources.
Data are Defined with Controlled Vocabularies	Data should be evaluated for attributes that carry different meanings to producers and users, and across disciplines.	• Promote metadata and data interoperability and reuse. Examples: ○ NERC Vocabulary Server ○ GCMD Keywords ○ CF Metadata Conventions
Data are Accessible ⁴	Data are most accessible when available in an immediately accessible form.	• Data are made available or published as soon as practicable. • Data with a URL in the metadata pointing to a single data file enables innovation systems to integrate with the data. • Data in non-standardised zip files, hidden behind passwords, or in online systems that restrict access are challenging for automated systems.

Data Repositories are Trustful ⁵	3rd party - repositories should be evaluated for their quality and carry certification ⁶ and qualities that data providers can check.	• Long term preservation needs to be guaranteed. • Data discovery promotes data legacy.
CARE Principles for Indigenous Data Governance ^{7,8}	Indigenous data should be treated to ensure that Indigenous governance over the data and its use are respected.	• The application of these principles should never compromise Maori Data Sovereignty.

References

1. SCAR Standing Committee on Antarctic Data Management: SCAR Data Policy. 2022. Accessed 3 May 2024. <https://scar.org/~documents/route%3A/download/5797>
2. The Antarctic Treaty. 1959. Retrieved from: https://documents.ats.aq/ats/treaty_original.pdf
3. Resolution 4 (1998) - ATCM XXII - CEP I, Tromsø. (1998). Secretariat of the Antarctic Treaty. Retrieved April 8, 2024, from <https://www.ats.aq/devAS/Meetings/Measure/258#>
4. Wilkinson, M., Dumontier, M., Aalbersberg, I. et al. The FAIR Guiding Principles for scientific data management and stewardship. Sci Data 3, 160018 (2016). <https://doi.org/10.1038/sdata.2016.18>
5. Lin, D., Crabtree, J., Dillo, I. et al. The TRUST Principles for digital repositories. Sci Data 7, 144 (2020). <https://doi.org/10.1038/s41597-020-0486-7>
6. CoreTrustSeal - Core Trustworth Data Repositories. CoreTrustSeal. 2024. Accessed 3 May 2024. <https://www.coretrustseal.org/>
7. Carroll, SR, et al. 2020. The CARE Principles for Indigenous Data Governance. mData Science Journal, 19: 43, pp. 1–12. DOI: <https://doi.org/10.5334/dsj-2020-043>
8. Royal Society Te Apārangi. Mana Raraunga Data Sovereignty. Royal Society of New Zealand Te Apārangi. 2023. <https://www.royalsociety.org.nz/assets/Mana-Raraunga-Data-Sovereignty-web-V1.pdf>

Related Links

- AntCat - New Zealand's Antarctic and Southern Ocean Metadata Catalogue. <https://antcat.antarcticanz.govt.nz/>
- AntCat Documentation - New Zealand's Antarctic and Southern Ocean Metadata Catalogue. https://antarcticanz.github.io/antcat_docs/intro.html
- Oceanum Datamesh. <https://docs.oceanum.io/datamesh/index.html>
- Polar Data Search. World Data System. <https://search.polder.info/>
- Antarctic Metadata Directory. NASA. <https://search.earthdata.nasa.gov/search?portal=amd>
- EMODnet Physics Product Catalogue. European Marine and Observation Data Network. <https://emodnet.ec.europa.eu/geonetwork>
- SOOSmap. Southern Ocean Observation System. <http://www.soosmap.aq/>
- OceanSites. EMODnet Endorsed Data Practices. <http://www.oceansites.org/data/>

Appendix

A.1. Recommended data archival formats, parameter naming vocabularies, and data repositories for different ASP data types formatted under this project.

Data type	Target Data Format	Parameter Vocabulary	Target Repository
Biodiversity	Darwin Core	Darwin Core	GBIF ¹

Raw Sequence	-	-	NCBI
Geophysical (e.g. seismic, ApRES – basal melt)	ASCII NetCDF	NERC ³ /CF ⁴ NERC ³ /CF ⁴	Pangaea
Atmospheric & Meteorology	NetCDF	NERC ³ /CF ⁴	Pangaea OSF ²
Oceanographic (e.g. moorings, CTD profiles)	NetCDF ASCII	NERC ³ /CF ⁴	Seanoë Pangaea
Ice – thickness, temperature, sipl etc	NetCDF ASCII	NERC ³ /CF ⁴	Seanoë Pangaea
Paleoclimate (e.g. grain size, elemental, diatoms)	ASCII Darwin Core	NERC ³ /CF ⁴ Darwin Core	Pangaea GBIF ¹
Modelling (e.g. model output)	NetCDF	NERC ³ /CF ⁴	OSF ²

1. [GBIF: Global Biodiversity Information Facility](#)
2. [Open Science Framework](#)
3. [CF Conventions Home Page](#)
4. [NVS Search](#)